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# Research Methodology
## Simulation Based Research

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## 1. Definitions

In this context the terms “modelling”, “computational modelling”, “numerical modelling”, and “simulation” are used interchangeably. Modelling is the use of computers to simulate and study the behaviour of complex systems using mathematics, physics, and computer science. The simulation results help researchers to make predictions about what will happen in the real system that is being studied in response to changing conditions for example. Models are helpful to analyse and understand complex systems for which an analytical or closed form solution does not exist.

## 2. Examples of Modelling

Some general examples of computational modelling are given below. Note that these are not necessarily engineering applications, and include medicine, biology, and financial systems.

- Climate, weather and solar activity
- Avalanche, earthquake, tsunami, and hurricanes
- Planetary and star movement
- Highway and street traffic
- Communications
- Electric power distribution, gas and petroleum pipelines
- Chemical, molecular, and subatomic particle behaviour
- Biological and medical processes
- Social behaviour
- Finances and investment market
- Supply chain logistics

## 3. Benefits of Modelling for Engineers

By using models, engineers can save time and money and can test systems and designs relatively risk-free. An (extreme) example is the testing of nuclear weapons which is physically dangerous. Another example is the testing of a motorised robot to collect dust from the Moon or Mars. Testing such a robot in reduced gravity and extreme temperatures would be difficult and expensive to execute on earth. The design and testing of deep-sea exploration equipment would also be time consuming and expensive if physical testing is to be undertaken. Modelling the system and performing a full structural analysis in this way, would be less expensive and safer. Modelling also offers the opportunity to analyse the effects of changing a single parameter or condition which is not always possible in physical testing since variables can be coupled.

Computational models can be used to fully optimise a design. For example, coupling the Finite Element Method (FEM) with an optimisation algorithm, the design parameters

(geometric dimensions, thickness, required material properties, etc.) can be optimised for a given load and boundary conditions.

Models provide the engineer with insight and data which are difficult, if not impossible, to obtain experimentally. For example, the flow inside an engine's intake manifold is complex and it is difficult to measure the whole velocity field. However, Computational Fluid Dynamics (CFD) can provide some insight.

Simulations can provide visual information (animations) which is helpful to understand the system, but it can also help to explain the process/system or product to non-technical and financial managers and be used for marketing purposes.

Models can be used to understand how a system will react under changing and non-ideal conditions. The risk associated with such changes can be quantified and the design made more robust (not overly sensitive to changing parameters/conditions, tolerances, etc.) if needed.

It is also important to remember that models are used to analyse complex systems for which a closed-form or analytical solution does not exist. Models are based on fundamental physics and mathematics, but some assumptions and simplifications are always needed and therefore a model can never be exact.

## 4. Examples of Engineering Modelling

Some examples of computational modelling approaches, often used by engineers, are given below. This includes FEM, CFD, Discrete Element Modelling (DEM), Rigid Body Dynamics (RBD), and control systems. If interested, follow the YouTube links provided for more details.

### 4.1 Finite Element Modelling

FEM is a computational technique used in engineering for solving complex problems related to solid mechanics, heat transfer, fluid dynamics, and other physical phenomena. It breaks down a large, complex structure or system into smaller, simpler parts called finite elements. These elements are analysed individually, and then their results are combined to predict the behaviour of the entire system.

FEM is widely used for structural and contact analysis (stress and strain, deformation, failure, static and dynamic loads, frequency response, fatigue, optimisation), fluid flow, thermal analysis, electromagnetic fields, and many other engineering applications, allowing for detailed, accurate predictions of performance and behaviour in complex systems that would be difficult or impossible to solve analytically.

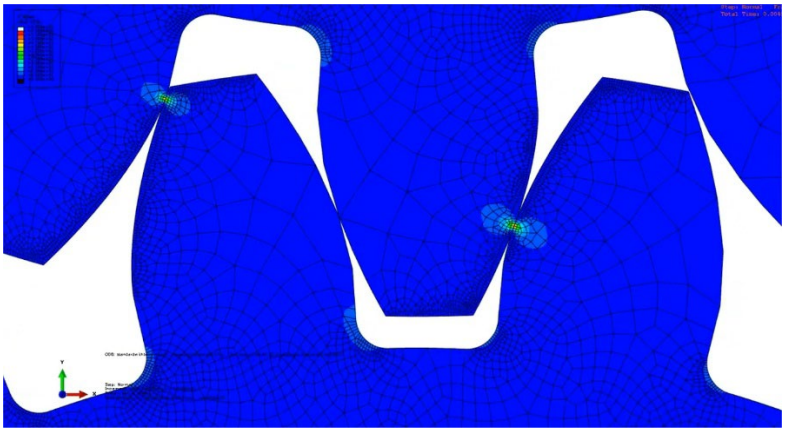

**Figure 1** – FEM model of a meshed gears in contact

### 4.2 Computational Fluid Dynamics

CFD is a branch of fluid mechanics that uses numerical methods and algorithms to analyse and simulate fluid flow, heat transfer, and related physical processes in various engineering applications. CFD allows engineers to predict how fluids (liquids and gases) will behave under different conditions without needing physical prototypes.

CFD is widely applied in various engineering fields, including:

- Aerospace engineering: Simulating airflows around aircraft, engines, and other components.
- Automotive engineering: To improve vehicle aerodynamics and engine performance.
- Civil engineering: Simulating airflow in buildings, bridges, and other structures, or studying environmental effects like pollution dispersion.
- Energy systems: In the design of heat exchangers, turbines, and air-conditioning systems.
- Chemical and process engineering: Modelling fluid flow in reactors, pipelines, and mixers.

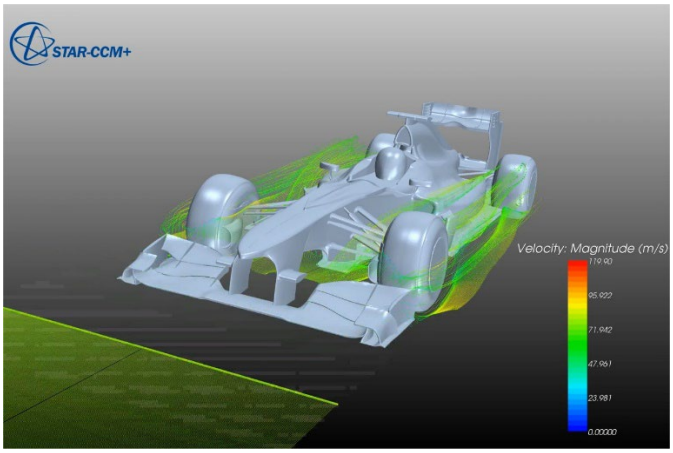


**Figure 2** – CFD model of the airflow over a racing car

### 4.3 Discrete Element Modelling

DEM is a computational technique used in engineering to simulate the behaviour of systems consisting of a large number of small, interacting particles or discrete elements. DEM is particularly useful for analysing bulk materials and particulate systems, where the individual interactions between particles (such as contact forces, friction, and collision) play a critical role in determining the overall behaviour of the system.

Key applications of DEM in engineering include:

- Bulk Material Handling - study the flow and behaviour of granular materials, such as sand, gravel, powders, or pellets, in systems like hoppers, conveyors, chutes, and silos. It helps to optimise designs to ensure smooth material flow, prevent blockages, and reduce wear.
- Mining and Quarrying - used to model the interactions between particles in mining equipment like crushers, mills, and screens. It aids in optimising equipment performance and improving energy efficiency in the crushing and grinding of ores.
- Geotechnical Engineering - modelling the interaction between granular soil particles and structural foundations or retaining walls. It is useful in studying the stability of slopes, landslides, or soil compaction and helps design foundations and underground structures.
- Agricultural Engineering - modelling the behaviour of crops (like grains or fruits) during harvesting and post-harvesting processes, such as sorting, handling, and packaging. It helps optimise machinery design to minimise damage to the produce. DEM helps simulate soil behaviour during tillage, helping to optimise plows and other agricultural equipment for better soil conditioning while reducing energy consumption.
- Pharmaceutical and Food Industries - DEM is used to simulate the behaviour of powder particles during processes like tablet compression or granulation in the pharmaceutical industry. It helps to improve the uniformity, strength, and quality of tablets and granules.

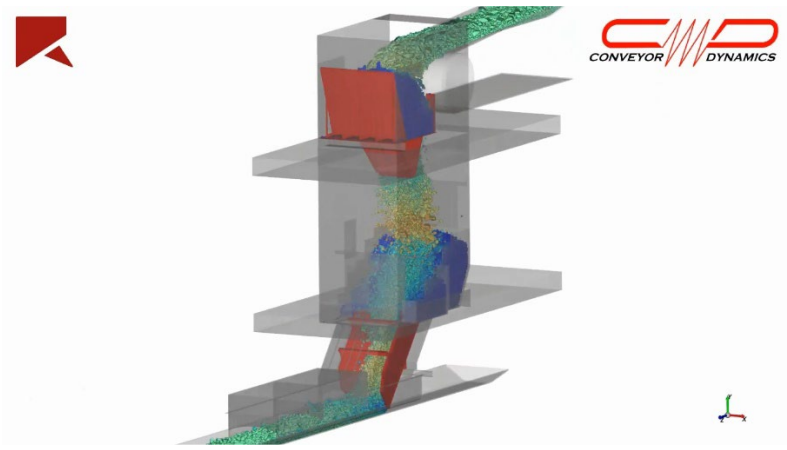


**Figure 3** – DEM model of a conveyor transfer chute

### 4.4 Rigid Body Dynamics

RBD in engineering refers to the study of the motion and forces acting on rigid bodies, which are objects that do not deform under stress. In other words, a rigid body is one where the distance between any two points within the body remains constant throughout its motion, even when subjected to external forces and torques. This simplification allows engineers to focus on the translation (linear movement) and rotation (angular movement) of objects without worrying about their internal deformations.

Applications of RBD in Engineering include:

- Mechanisms and Machines - analysing the movement of components in mechanical systems, such as gears, linkages, cams, and robotic arms. Understanding the motion and forces involved allows engineers to design systems that function efficiently and reliably.
- Vehicle Dynamics - in automotive, aerospace, and marine engineering, rigid body dynamics is used to analyse the motion of vehicles and spacecraft, including their acceleration, braking, turning, and stability. It helps engineers optimise the design of vehicles for performance, safety, and fuel efficiency.
- Robotics - the study of rigid body dynamics is fundamental in the design and control of robots. The movement of each segment or link of a robot manipulator can be analysed as a rigid body, helping to develop efficient motion control strategies and optimize design for load-bearing and precision.
- Biomedical engineering - rigid body dynamics plays a critical role in understanding and analysing the motion and forces applied to biological systems, medical devices, and prosthetics. Even though biological systems often involve soft tissues that deform, there are many instances where components can be treated as rigid bodies for simplification and effective analysis.

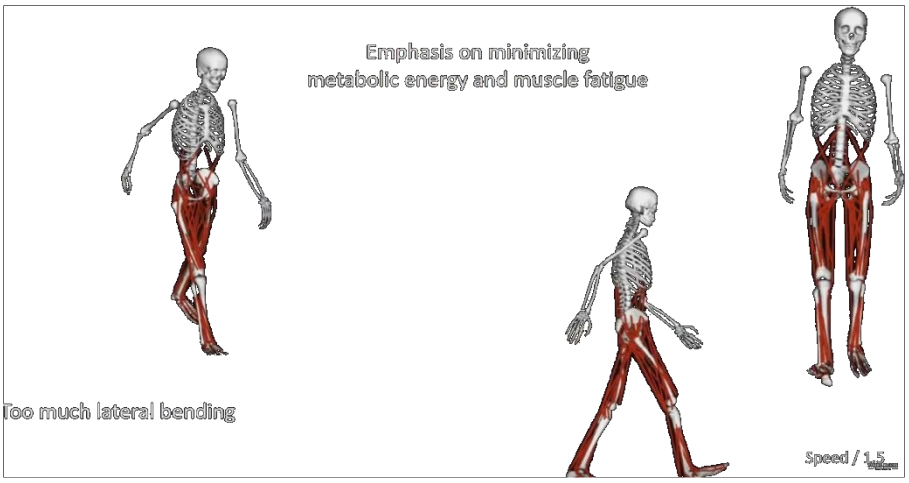


**Figure 4** – RBD model in biomedical engineering

### 4.5 Control System Modelling

Control system modelling in engineering refers to the process of developing mathematical representations or models of a system to predict its behaviour and to design controllers that can regulate its performance. These models describe how the system responds to inputs, disturbances, and feedback, helping engineers design systems that maintain desired outputs or behaviours despite uncertainties, disturbances, or changing conditions.

By creating accurate models of systems, engineers can design control strategies that ensure stability, performance, and reliability across a wide range of applications, from mechanical systems to electrical, thermal, fluid, and biomedical systems. Whether through transfer functions, state-space models, or numerical simulations, control system modelling is essential for modern engineering design and analysis.

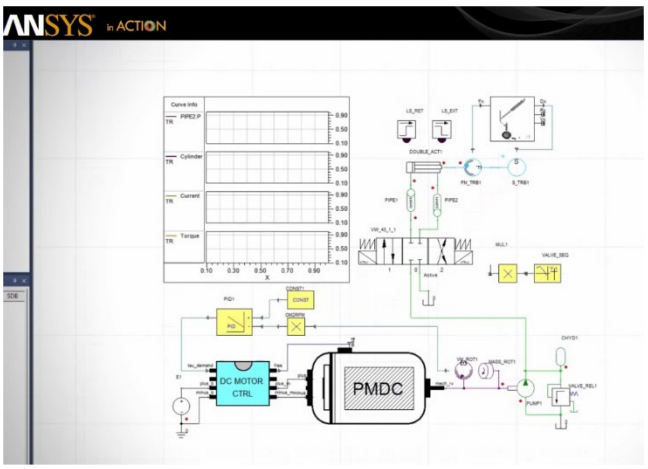


**Figure 5** –An example of a control system model which includes the hardware

### 4.6 Multi-Physics Modelling

Multi-physics modelling is the process of creating a computational model that simulates the behaviour of a system where multiple physical phenomena interact with each other. These phenomena can include, but are not limited to, mechanical, thermal, electrical, fluidic, and chemical processes. Multi-physics models are used to understand complex systems where different physical effects influence one another simultaneously, providing a more accurate and holistic understanding of real-world behaviours compared to models focusing on a single physical domain.

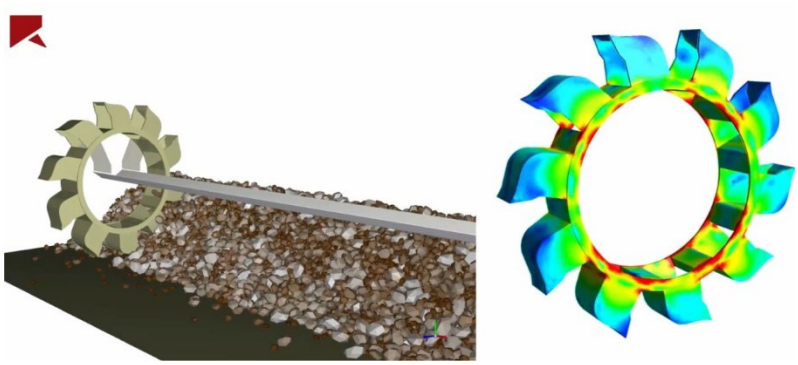

**Figure 6** –Multi-physics model of a bucket wheel excavator using DEM and FEM

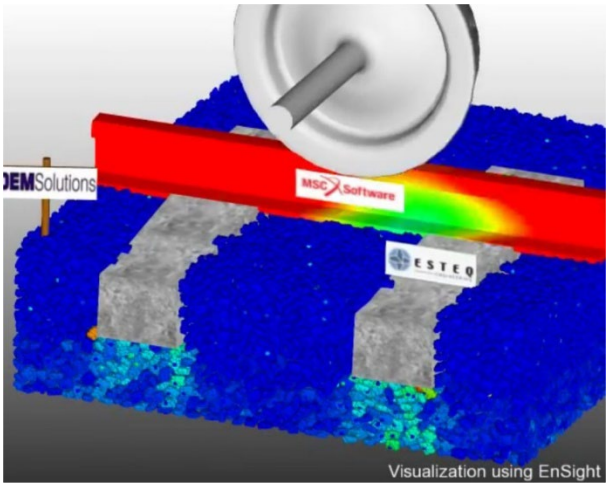


**Figure 7** –Multi-physics model of a railway track using DEM and FEM

**Figure 8** –Multi-physics model of a fluidised bed using DEM and CFD

## 5. Research and Modelling

Using computational modelling for research purposes is different from using it as a design tool. Engineers in practice would often make use of models or simulations as a design tool, and although the same basic steps, such as verification and validation, are important, they are often ignored. It is never good practice to use a simulation model as a "black box" without understanding the fundamental principles, assumptions, simplifications and limitations.

In a research environment however, models and simulations can in general be used in three different ways: the researcher can either (1) develop a completely new numerical model or simulation software, (2) develop a sub-model which will link with an existing (often more complex) model or simulation software, or (3) use available models or simulation software to analyse a specific system or application.

In the first two cases, the research focus is on developing the new model or sub-model. New models are developed by deriving the theory or mathematical model from scratch and implementing the computer code (software). Developing sub-models might include for example a new type of element for FEM, which is then integrated or coupled to existing FEM models and software, or the coupling of a newly developed optimisation model to existing FEM, CFD, DEM or RBD models and software.

In the third case, the use of a model or simulation is very similar to that of an engineer in practise using it as a design tool, however, the focus would not be solely on designing a functional system, but rather to gain a better understanding (research) of the fundamental system behaviour, optimising the system, analysing alternatives, varying conditions and input parameters, etc. Here the focus is not on the fundamental theory and implementation of the model, but rather the correct use of the model or software for the specific application. It is still very important for the researcher to understand the fundamental principles, assumptions, simplifications and limitations of the selected computational model/software.

## 6. Modelling Methodology

Figure 9 proposes a methodology for computational modelling in a research environment, which is applicable if a new model, a sub-model or existing models or simulation software is used. The model development runs parallel with the experimental development which is essential for validation purposes. Each step in the process is explained below.

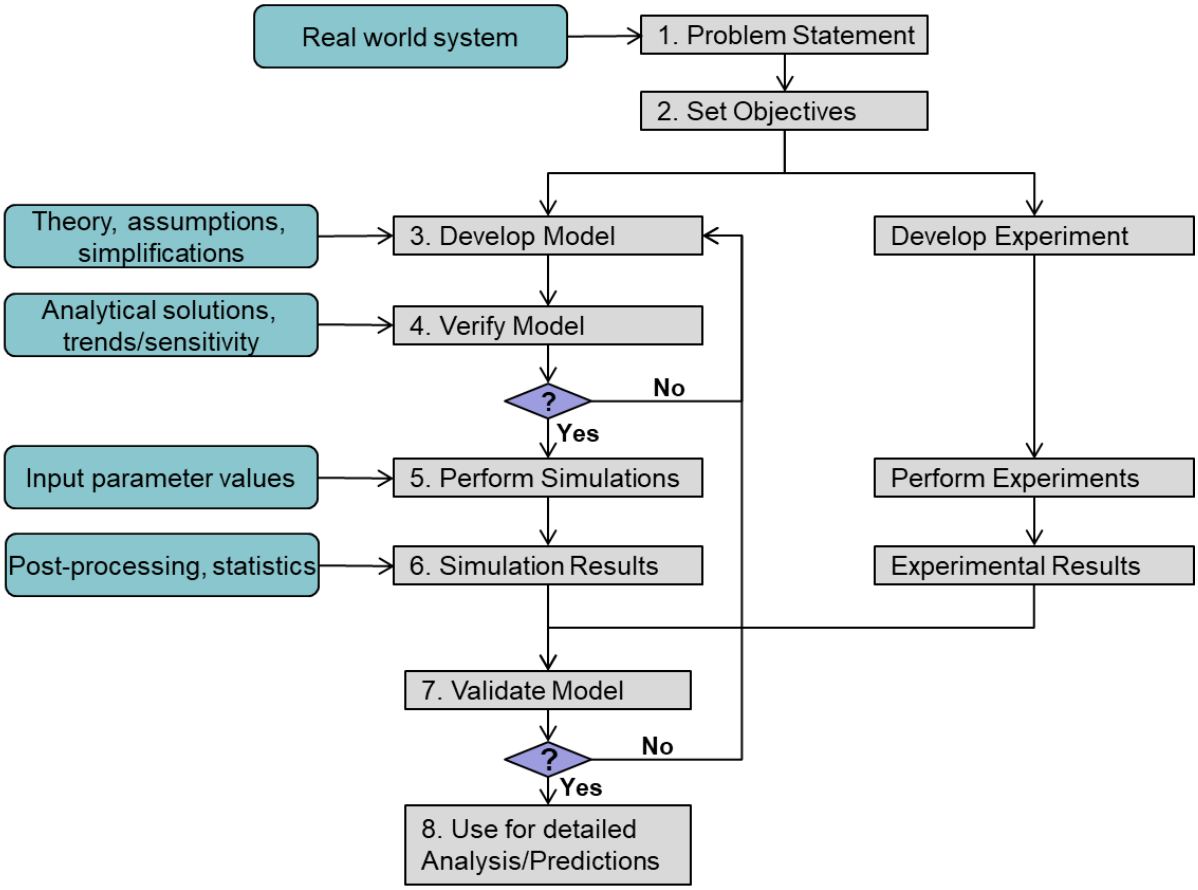


**Figure 9** – Modelling methodology flow diagram

### 6.1 Problem Statement

It is very important to have a clear problem statement before developing the model. A model is always a simplified replica of a real-world system where fundamental physics are at play. State and describe the real-world system, identify the physics that need to be modelled (a multi-physics approach might be needed) and base or select the computational model on this.

### 6.2 Objectives

List all the objectives of the research and specifically those of the modelling aspect at an early stage. Also consider the measures that will be used to verify and validate the model. Determine the accuracy required when the results of the model are verified or validated. In FEM for example, the strain must be within 10% and the displacement within 5% of the measured value (this can be application dependent). Even if existing software is used, the objectives will influence the required accuracy, the data to record/export, simplifications, etc.

### 6.3 Develop the Model

If an existing model or software is used, this step can include creating of the system geometry, setting the boundary/initial conditions, setting material properties, selecting the parameters and sub-models to use. In general, modelling the real-world system using existing software can include one or more of the following activities:

- Select appropriate and reliable software.
- Understanding the fundamental physics and theory is important.
- Consider any assumptions and simplifications.
- Select the correct type of model, for example, linear versus non-linear FEM, laminar or turbulent flow, etc.
- Keep in mind the objectives.
- Model the real-world system using existing software.

If a new model, sub-model or simulation software is developed, it would include a careful analysis of the fundamental physics involved, assumptions and simplifications that are needed and appropriate, deriving of a mathematical model and finally coding of the model. In general, this step can include one or more of the following activities:

- Based on fundamental theory and physics, develop a mathematical model to represent the real-world system.
    - Consider any assumptions and simplifications.
    - Keep in mind the objectives.
- Develop computer code/software based on the mathematical model developed above. Make use of computer science/programming skills.
- Model the real-world system using existing software.

When working with a large or complex model, it is useful to divide it into sub-models. Each sub-model can be individually verified and validated as discussed later. Working with sub-models make it easier to debug the code/model and easier to understand how specific parameters influence the model.

### 6.4 Verify the Model or Sub-Models

Verification of models and/or sub-models is a very important step. Verification is the process of determining that the model operates as intended and can include debugging of the code/model. In general, the verification step can include one or more of the following:

- Compare the model results to analytical results or known and trusted results from another source (for simplified models).

- Check if the trends are correct when changing the input. For example, when increasing the heat flux, the temperature should increase. Check for unexpected discontinuities.
- Check if the model is sensitive to all input variables. When a single input variable is changed, the system response should change. The level of sensitivity might not be important.
- Even if commercial FEM/CFD/DEM or other software is used to model a specific application, some form of verification should still be conducted, for example convergence on mesh/cell refinement.
- Verify each of the major components of the model.
- Complex models might never be absolutely verified!

### 6.5 Perform Modelling for Validation

In this step, the model is used to analyse a real-world system for validation purposes. Usually this would be a model of an experiment where the input variables can be accurately set, and the output variable accurately measured. The validation experiment should either be the final system/application to be analysed, or very similar to it in nature. If the validation experiment is vastly different from the final system/application to be analysed, it is possible that the model could be successfully validated, but this provides no guarantee that it will also accurately model the final system/application. Keeping the validation experiment/model closely related to the final system/application, one can expect the model to be accurate.

The model geometry, boundary conditions, initial conditions, material properties, etc. would be based on those used in the validation experiment. This model should replicate the physical experiment as closely as possible, considering the assumptions and simplifications which are unavoidable but acceptable.

### 6.6 Modelling Results

Carefully analyse the results. Make sure to work with the correct results/data and use appropriate tools for plotting and statistical analyses. Quit often, the results do not seem as expected, but this does not necessarily indicate that the model is wrong. First make sure that the correct data is analysed and that the post-processing calculations are correctly performed. If the answers are out by a factor of 10, make sure the input variable had the correct units, for example.

### 6.6 Validate the Model

Validation is the process of ascertaining that the model is an acceptable representation of the real-world problem and instils confidence in the model. This is done by comparing the simulation results to experimental results of the same system or other known and trusted results. Compare the errors with what is expected of such a model. Consult

literature to estimate what an acceptable error margin is for this type of analysis. If the model is not successfully validated, the fundamental physics, assumptions and simplifications and mathematical model should be revised as needed.

If existing software is used, the model should be carefully checked: are the correct sub-models used, are the material models, element type, boundary layer model, boundary conditions, etc., appropriate.

### 6.7 Use the Model for Detailed Analysis/Predictions

With a successfully validated model, the final system/application can be analysed. Often the validation experiment/model is the final application of interest. If this is the case, variables that can be accurately measured in the experiment would be used for validation, for example the inlet and outlet flow of a manifold. In this final step, the model can then be used to analyse the internal flow, which cannot be measured.

On the other hand, the validation experiment/model can be closely related to the final system/application to be analysed but reduced in size and complexity. If this is the case, a model of the final system/application should be created.

This final step is where progress and new discoveries are made, and novel ideas are born. The model can be used to understand the fundamental system behaviour, to predict how the system would respond to different design and input parameters. The model can be coupled with optimisation algorithms to obtain the optimal design/settings.

## 7. Example of Computational Modelling

An example of using CFD in a research project is presented [1]. Sub-models are created for verification and validation purposes based on an existing design. With a fully validated model, a new design is proposed, analysed and shown to perform better than the existing standard design.

### 7.1 Problem Statement and Objectives

**Problem statement:** Optimisation of multi-scale ventilated package design for next-generation cold chain strategies of horticultural produce.

**Physics involved:** The flow of cold air through packaged fruit inside a refrigerated container. Heat transfer between the fruit and the cooling air. See Figure 10 for details on the different levels of packaging.

**Objectives:**

- Develop a validated computational model capable of predicting transfer processes (airflow and heat) in multi-scale structured horticultural packaging systems.
- Apply modelling to analyse the performance of cooling operations and packaging design using a multi-parameter approach.

- Based on the insights gained from the above objectives, design next-generation optimal multi-scale ventilated packaging for postharvest handling of fresh produce, under improved utilisation of refrigerated freight container space.

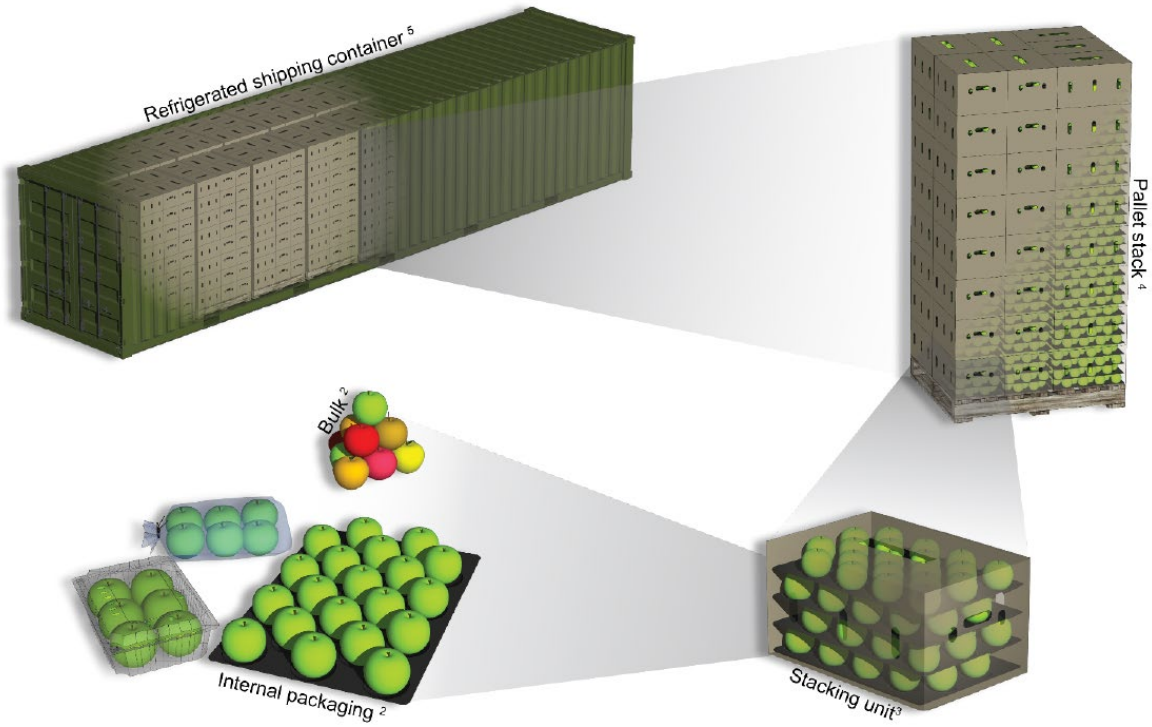


**Figure 10** – Example of a standard box design, filled with fruit, and loaded into a refrigerated shipping container

### 7.2 Develop, Verify and Validate the Model

This research clearly needs to make use of CFD software and does not require the development of new models or simulation software. Existing software can be used, and in this case ANSYS Fluent was selected.

The modelling is divided into a number of sub-systems where each can be verified and validated. A CFD model is developed for each sub-system by accurately modelling the geometry and boundary conditions.

The first sub-model is that of a single existing standard box filled with fruit as shown in Figure 11. Boundary conditions are set similar to that of an experiment where a single box with fruit is fitted inside a wind tunnel. Appropriate turbulence and boundary layer models are selected.

The sub-model is verified by estimating the pressure loss across the standard box from fundamental flow theory and comparing it to model results. The sub-model sensitivity is analysed in terms of the inlet velocity (flow rate), temperature and the pressure loss across the box.

The sub-model is validated by comparing the cooling time of the fruit with measurements for different vent hole designs (Figure 12). Further validation includes comparing the pressure drop across the box as a function of the flow rate with the measurements (various vent hole designs).

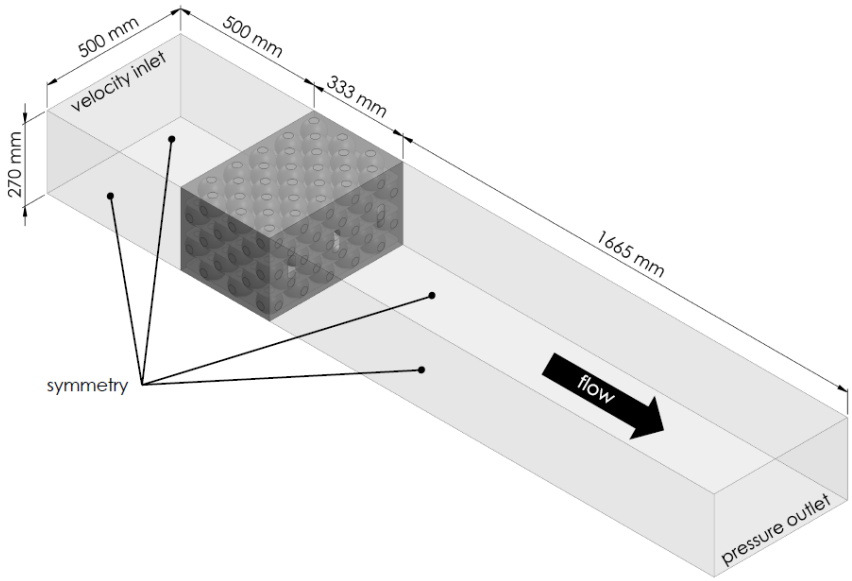


**Figure 11** – A CFD sub-model of a single box filled with fruit and placed inside a virtual wind tunnel with a specified airflow

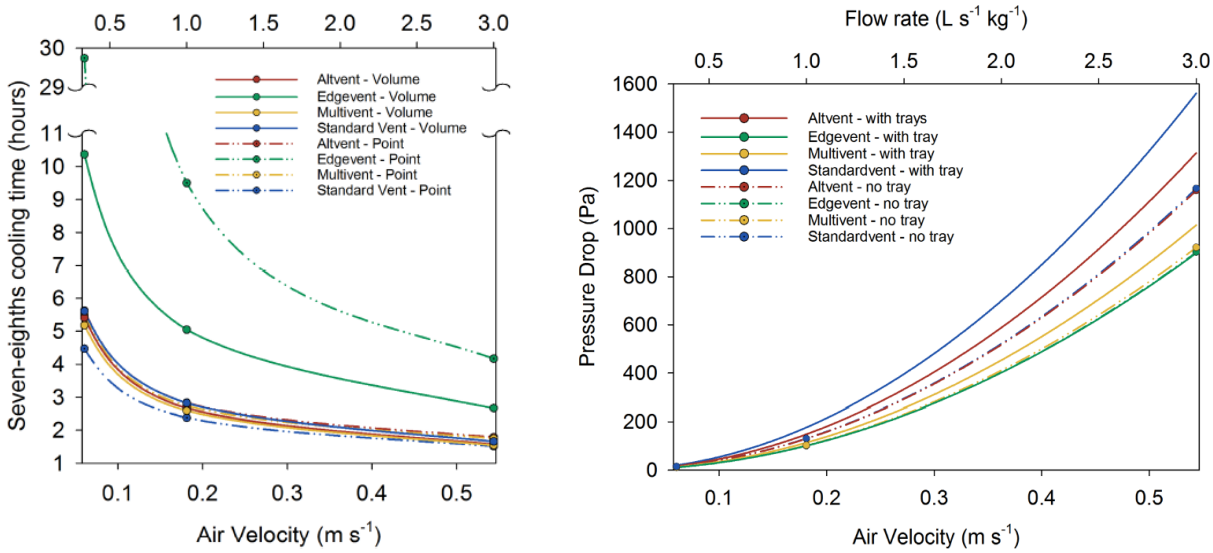

**Figure 12** – CFD sub-model validation

Building on the first sub-model where a single box is analysed, the second sub-model is also that of the standard (existing) box design, but with multiple boxes on a palletised row with only the horizontal flow analysed, Figure 13. The model is validated against experimental measurements of pressure loss versus flow rate and also the cooling rate of the fruit.

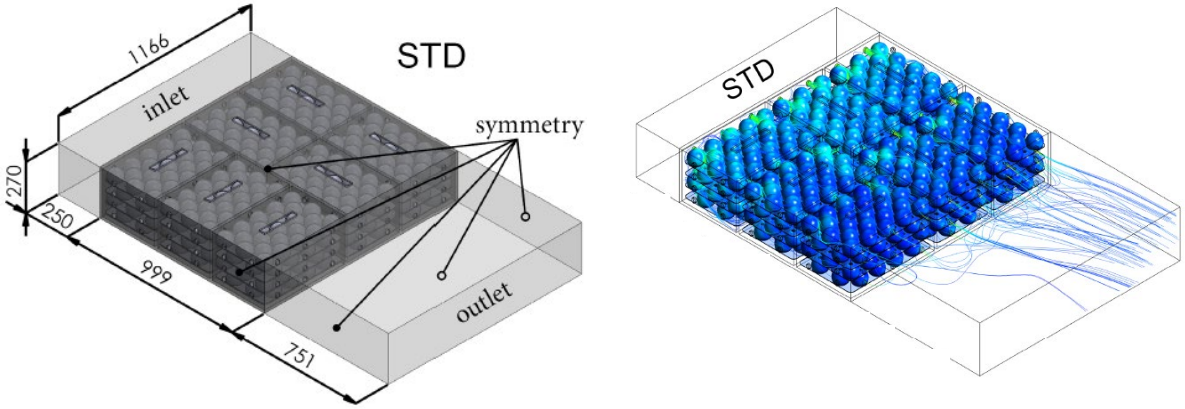


**Figure 13** – A CFD sub-model of a single row of boxes filled with fruit and placed inside a virtual wind tunnel with a specified airflow

The third sub-model is also that of the standard (existing) box design, but multiple boxes are stacked on a pallet and the vertical flow analysed, Figure 14. The model is validated against experimental measurements of pressure loss versus flow rate and also the cooling rate of the fruit.

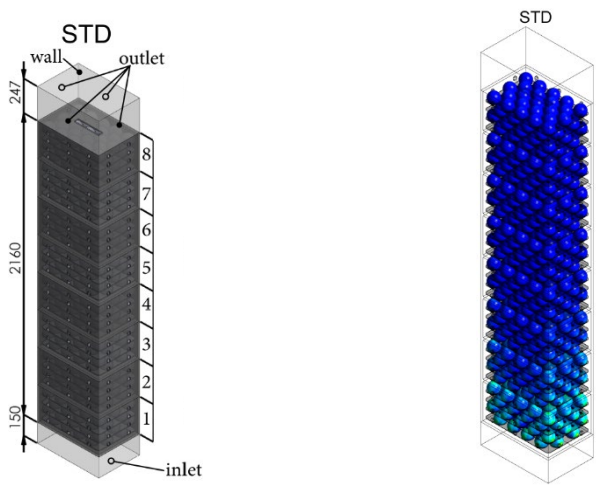


**Figure 14** – A CFD sub-model of a single column of boxes filled with fruit and placed inside a virtual wind tunnel with a specified airflow

### 7.3 Using the Model to Design a New Box (Prediction)

A detailed analyses of the standard design in the previous steps, showed its shortcomings and how it can be improved. With a thoroughly validated model, a completely new box designs can be proposed and its performance analysed without the need for any experiments, Figure 15. The new design can be optimised to reduce pressure loss (requires less energy) and increasing fruit cooling efficiency.

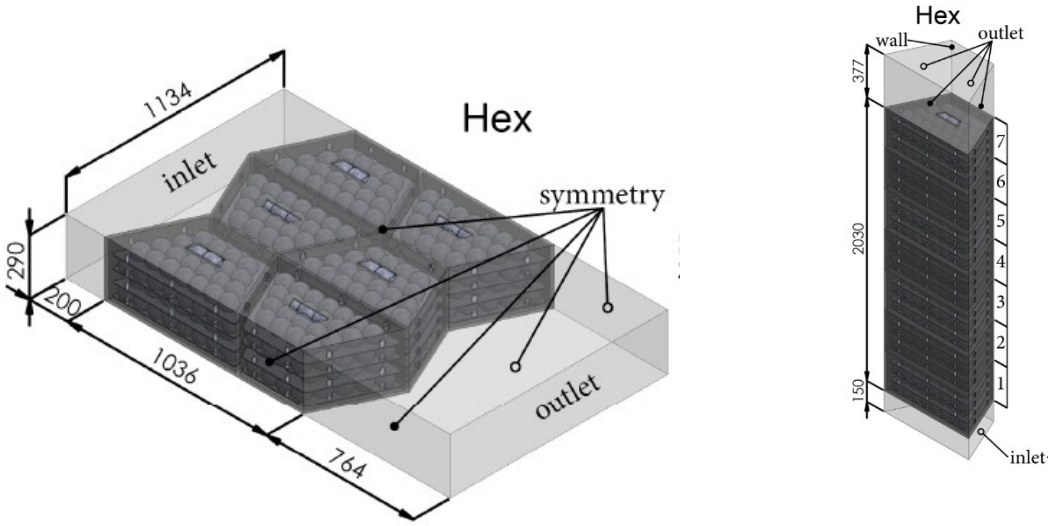


**Figure 15** – CFD models of a newly designed box and palletisation system

The performance of the new design can be compared to that of the standard design in ways which are difficult, if not impossible, to do experimentally (Figure 16). This provides further insight into the fundamental behaviour of the system.

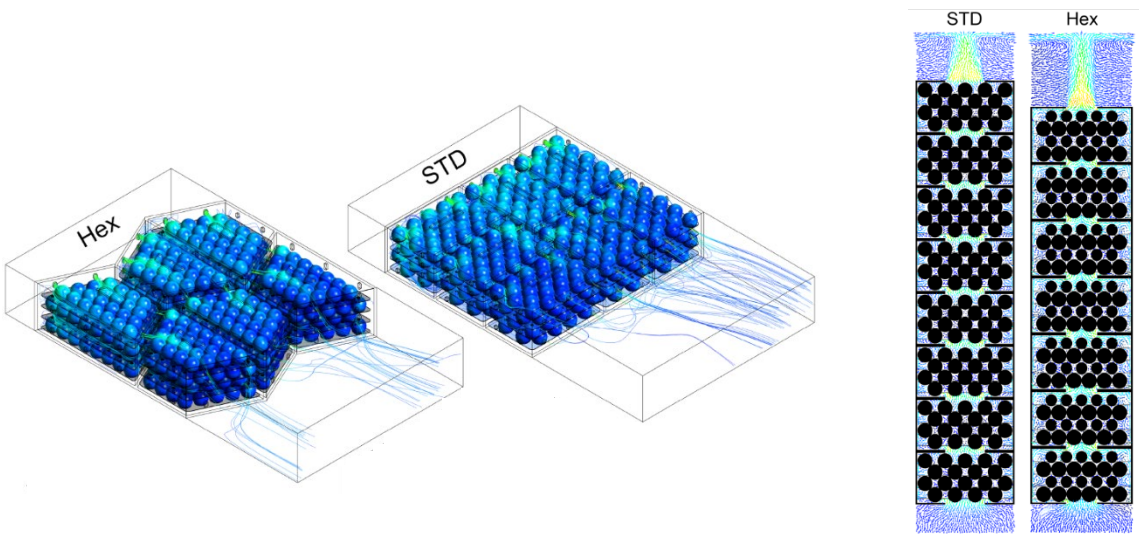


**Figure 16** – CFD models comparing the old and new designs

## References

[1] Berry, T, 2017, Optimisation of multi-scale ventilated package design for next-generation cold chain strategies of horticultural produce, PhD Thesis, Department of Mechanical and Mechatronic Engineering, Stellenbosch University